MNL Nash Competition

Richard P. Rumelt

Exhibit 1 displays data on three competing products—the buyers' values (w_k), the producer unit costs (c_k), and the maximum potential surplus created by each ($w_k - c_k$). If competition were intense, product 2 would win because it is capable of offering more surplus than the other two. At a price of 9.99 it offers 4.01 of surplus to the buyer, more than either competitor can.

What if competition is present but not that fierce? The MNL model provides just such a setting—one in which a larger offering of surplus garners more market share but does not completely dominate. In this note we explore the Nash equilibrium in prices implied by an MNL market model.

Exhibit 1—Three Competing Products (Again)

	Product 1	Product 2	Product 3
$\mu = 1$			
WTP (w)	12.00	14.00	16.00
Unit Cost (c)	8.00	9.00	13.00
Potential Surplus	4.00	5.00	3.00

The Nash equilibrium is a set of prices (p_k) such that no producer, acting alone, has an incentive to change their price. There is no simple formula yielding these prices. They must be computed by a form of iterative calculation. In essence, each product solves an independent optimization problem.

To see how this works, suppose that the three products begin with prices set to (10, 12, 14}. These prices are each 2 below the WTP for that product. With equal amounts of surplus for each product, each product will have a market share of 1/3. Now, would product 1 be better off with a different price?

In Excel, one can use Solver to find the price which maximizes product 1's profit. To follow along this exercise, load worksheet Nash_ex1.xlsm. It has built-in buttons to run Solver and/or Premium Solver on each product. It also has a built-in function that computes MNL market shares.

Invoking Solver on product 1, p_1 falls from 10 to 9.69, and 1's share rises from 33% to 41%. Invoking Solver again for product 2, we find its price falls from 12 to 11.07. Given those two price drops, product 3 drops its price from 16 to 14.21. Returning to product 1, we find a second drop from 9.69 to 9.49. Repeating this process until there are no further changes gives the Nash

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Exhibit 2—Three Competing Products: Nash Equilibrium

	Product 1	Product 2	Product 3
WTP (w)	12.00	14.00	16.00
Price (p)	9.49	11.04	14.19
Buyer Surplus (b)	2.51	2.96	1.81
Unit Cost (c)	8.00	9.00	13.00
Unit Profit	1.49	2.04	1.19
Gross Margin	16%	19%	8%
Market Share	32.8%	51.1%	16.2%
Gross Profit	487.24	1,043.43	192.97

equilibrium. In this case it is shown in Exhibit 2. (Here, the "gross profit" has been calculated as if there were 1000 buyers. This scaling is immaterial unless there is more than one segment.)

There are no commercial products for delivering this result. We can, however, trick Solver into finding this solution in one step—if there aren't too many products or complexities. A product's price is optimum when the partial derivative of profit with respect to price is zero. In the MNL model, this corresponds to the condition that

$$1 + m_k E_{q,p} = 0 \tag{1}$$

where

$$E_{S_k, p_k} = -\frac{p_k}{\mu}(1 - S_k) \tag{2}$$

What (1) says is that a price is optimum when the profit margin is the negative inverse of the price elasticity. It is a simple matter to add the computation of (1) for each product to our worksheet. Restoring the original trial prices of {10, 12. 14}, we find that the value of this quantity is {-.33, -1, and 0.33} for the three products. Define a cell SSE which contains the sum of these three squared "errors." Then ask Solver to minimize this value (e.g., make it zero) by setting the three prices. You should get the Nash solution in one step.

Outside Values

In the example just solved, the buyers only had three choices: product 1, product 2, or product 3. Often, it is useful to specify a generic alternative to the competing products—an outside alternative to buying anything in this category. To do so, we create a fourth virtual product with value *Vout* and a price of zero. Choosing this alternative provides a buyer surplus of *Vout*.

The worksheet Nash_ex1.xlsm includes a variable *Vout* which is linked to the user-defined function MNL1Shares. This function takes arguments *rUnitPrice*, *Mu*, *Vout*, and *rWTP*. The arguments with "*r*" prefixes take a range of cells as an argument, while *Mu* and *Vout* take single values. The function returns the shares of the products as an array—you must select the row of cells to report within and use CTRL-SHIFT-Enter to mark the formula as an array formula.

Exhibit 3—Three Competing Products: Nash Equilibrium with Outside Good

	Product 1	Product 2	Product 3	Totals
WTP	12.00	14.00	16.00	
Unit Price	9.28	10.67	14.11	
Buyer Surplus	2.72	3.33	1.89	
Unit Cost	8.00	9.00	13.00	
Unit Profit	1.28	1.67	1.11	
Gross Margin	14%	16%	8%	
Buyer Share	21.7%	40.0%	9.5%	71%
Market Share	30.5%	56.2%	13.3%	100%
Units Sold	217.46	400.33	95.09	
Gross Profit	277.89	667.60	105.08	

Note: μ=1, *Vout* = 3, 1000 potential buyers.

When *Vout* is set to a negative value, it is ignored. If it is set to a non-negative value (including 0), shares are adjusted to reflect some buyers choosing the outside good. In this case, it is useful to distinguish between *buyer shares* and *market shares*. A buyer share is the share of all buyers choosing a product. A market share is the share of all buyers not choosing the outside good who choose a product.

Worksheet Nash_ex3.xlsm displays the problem we have been studying with *Vout* = 3. Click the OneStepSolver button and see the solution. The result is shown in Exhibit 3. Prices fall a bit and the total buyer share for all three product falls to 71%. That is, 29% of buyers chose the outside good. Profits are down substantially with this new competition reducing the total number of buyers.

Multiple Segments

The analysis of product competition becomes especially complex and interesting when there is more than one market segment. To model this, we apply an MNL model to each segment, letting each segment have unique values for μ, *Vout*, and for *QSeg,* the number of buyers in the segment. In addition, each product has a different set of willingness-to-pays for each segment.

Worksheet Nash_ex4.xslm extends the example problem by adding a second segment. The new segment has four times the number of buyers and these buyers are highly prices sensitive (μ = 0.5). And, they do not appreciate quality differences and assign a WTP of 12 to each product.

Each product adjusts its price to maximize profits summed across both segments. The result is displayed in Exhibit 4. Comparing these results to Exhibit 2, product 1's price has risen along with its market share. Product 2's price has fallen along with its market share. Product 3's has risen and its results are much worse.

In the original market segment, product 2 had the competitive advantage (refer to the "Potential Surplus" line in Exhibit 4.) In the new segment, product 1 has the advantage and product 3 is at a large disadvantage. [More to come later]

Exhibit 4—Three Competing Products: Nash Equilibrium with Outside Good

	Product 1	Product 2	Product 3
WTP	12.00	14.00	16.00
	12.00	12.00	12.00
Unit Price	9.62	10.04	14.10
Unit Cost	8.00	9.00	13.00
Gross Margin	17%	10%	8%
Buyer Share	15.4%	75.0%	9.6%
	69.9%	30.1%	0.0%
Units Sold	2,948.47	1,955.30	96.23
Overall Market Share	59.0%	39.1%	1.9%
Gross Profit	4,788.83	2,042.14	105.96
Potential Surplus	4.00	**5.00**	3.00
	4.00	3.00	(1.00)
Delivered Surplus	2.38	**3.96**	1.90
	2.38	1.96	(2.10)

Note: μ=1, *Vout* = 3, 1000 potential buyers.